

EcoPulse

Green Radar Systems for Climate-Resilient Flood Monitoring

AESS Sustainability Hackathon 2026 ð Phase 1 Submission

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IEEE SUTech Student Branch ð AESS SUTech SC

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Project title	EcoPulse
Team name	Elsewedy University of Technology
Student Branch	IEEE SUTech SB, AESS SUTech SC
Problem area	Flash floods in arid wadis Egypt's most destructive natural hazard
Proposed solution	Sustainable dualmode 24 GHz FMCW radar with AI fusion, blockchain security, and solaronly autonomy
Core proof	Endtoend MATLAB/Octave simulation (42 modules) validated across 5 flood scenarios and a 20run Monte Carlo evaluation
Tools used	MATLAB, Octave (toolboxfree), Python (energy model)

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1 Project Summary

EcoPulse redefines flood early warning by merging green radar technology with explainable AI. The project demonstrates that a lifesaving early warning system can operate entirely on solar power, run on free software, and be deployed for less than the cost of a single flood-damaged bridge. This concept document outlines the problem, solution, technical design, simulation results, and the project's alignment with sustainable development goals.

2 Problem Statement

2.1 What is the Problem?

Flash floods in Egypt's arid wadis (dry river valleys) strike with devastating speed, often with no warning. Water levels can rise by several meters within minutes, sweeping away people, livestock, and infrastructure. Traditional instream sensors fail immediately upon debris impact, and commercial radar systems are too powerhungry for remote, off-grid locations. Currently, **no integrated lowpower, AI-augmented early warning radar network exists** over Egypt's most vulnerable wadis.

2.2 Who is Affected?

- **Rural communities** in Sinai, Upper Egypt, and the Red Sea Hills living directly in flood paths.
- **Urban centers** (Cairo, Alexandria) where inadequate drainage causes repeated inundation.
- **Critical national infrastructure:** highways, power grids, oil pipelines, agricultural land.
- **National economy:** billions of Egyptian pounds spent yearly on emergency response and reconstruction.

2.3 Why Does It Matter Now?

- **Climate change** is intensifying extreme rainfall events across North Africa floods are becoming more frequent and severe.
- Egypt has a **national adaptation strategy** and a **\$7B World Bank commitment** (through 2027) for water resilience, but lacks the appropriate sensing technology for remote wadi monitoring.
- Every year without autonomous early warning means **preventable deaths, repeated infrastructure destruction, and mounting economic losses**.

2.4 RealWorld Impact in Egypt

- **March 2020 "Dragon Storm":** 21+ dead, heaviest rainfall in 40 years.
- **2020 floods:** 40+ killed, 400+ injured across multiple events.
- **2016 floods:** 32,500 people directly affected, homes submerged within hours.
- **Alexandria by 2050:** a projected 30% of the city flooded, 1.5 million displaced, 195,000 jobs lost.

These are not projections they are accelerating realities.

3 Proposed Solution

EcoPulse is a **selfcontained, AIaugmented 24 GHz FMCW radar system** that provides realtime, noncontact waterlevel monitoring and early warning for flash floods in remote wadis.

3.1 How It Works In Simple Terms

1. **Send & Listen:** A radar on a tower sends invisible radio pulses downward and listens for the echo bouncing off the water surface.
2. **Measure Distance:** The echos roundtrip time gives the exact water level updated every second.
3. **Watch the Trend:** The system tracks both the current level and how fast it is rising.
4. **Dont Panic Verify:** If a threshold is crossed, the radar waits **60 seconds** to confirm the trend is real (not wind, debris, or a passing truck).
5. **Confirm & Act:** Only then does it switch to highaccuracy mode, trigger the alarm, log the event securely, send warnings (SMS/Telegram/LoRa), and compute the safest evacuation route.

3.2 What Makes EcoPulse Different

- **Green dualmode radar** lowpower ZadoffChu for continuous monitoring, highresolution Chirp only for verified threats (used <5% of the time) months of solaronly operation.
- **Multievidence AI fusion** four independent decision engines (rulebased, fuzzy logic, Bayesian SPRT, ARIMA forecast) must agree before an alarm is raised; a SHAPbased XAI module explains every decision.
- **Complete resilient ecosystem** tamperproof blockchain ledger, cyberdefense, DEMbased evacuation routing, 3D digital twin, multichannel alerts (SMS/Telegram/LoRa).
- **Toolboxfree & open** all 42 modules handcoded in MATLAB/Octave, zero license barriers, ready for lowcost hardware.
- **Purposebuilt for remote wadis** designed from scratch for Egypts harsh, offgrid valleys, not an offtheshelf adaptation.

4 System Architecture

EcoPulse consists of **42 files** organized into three layers:

4.1 Core Pipeline (25 files)

Waveform generation (ZadoffChu & Chirp), environmental model (PiersonMoskowitz waves, ITUR rain attenuation, multipath), signal processing (matched filter, CFAR, Kalman tracker, MUSIC superresolution), feature extraction, decision logic, and alarm validation.

4.2 Professional Modules (15 files)

AI/ML (LSTM surrogate, online anomaly detector, fuzzy/Bayesian/ARIMA fusion, XAI), IoT (sensor network, alert gateway), security (blockchain ledger, cyberdefense), GIS

(DEM evacuation), energy (solar manager), and AR/VR export.

4.3 Ultimate GUI (1 file + launcher)

A 7tab interactive dashboard with live waterlevel charts, 3D digital twin, rangeDoppler map, risk gauge, energy monitor, event log, and full parameter control.

4.4 DataFlow Summary

Sensors Radar echo + weather data
Processing Matched filter CFAR Kalman Feature extraction
Fusion Rule + Fuzzy + Bayes + ARIMA Risk score
Validator 60sec NoutofM persistence check
Actions Alert gateway (SMS/Telegram/LoRa), blockchain log, evacuation route, 3D twin update

5 Technical Implementation

5.1 SoftwareDefined Radar Specifications

- **Waveforms:** ZadoffChu ($N = 127$, $u = 5$) for monitoring; LFM chirp ($B = 20$ MHz, $T_p = 20 \mu\text{s}$) for alert.
- **Processing chain:** matched filter, CA/OSCFAR ($P_{fa} = 10^{-4}$), Kalman CV tracker, ARIMA(1,1,1) forecaster, handrolled LSTM cell.
- **Environmental model:** PiersonMoskowitz wave spectrum, ITUR rain attenuation, multipath, colored noise.
- **Energy model:** 30 W solar panel, 50 Wh battery highpower Chirp mode used <5% of simulation time.

5.2 AI & Algorithms (All HandCoded, ToolboxFree)

- Multievidence fusion: rulebased thresholds, Mamdani fuzzy inference, Bayesian Wald SPRT, ARIMA forecast.
- LSTMstyle surrogate forecaster (scalar gates), online anomaly detector (Welfords zscore).
- SHAPbased XAI explainer providing featurecontribution ranking for every alarm.
- Edge AI (TinyML int8) simulated for microcontroller portability.

5.3 Simulation Prototype

EcoPulse is currently a **fully functional, endtoend MATLAB/Octave simulation**. All 42 modules have been rigorously tested across multiple flood scenarios and a Monte Carlo evaluation. The codebase is entirely toolboxfree, ensuring seamless future deployment on lowcost SDR hardware.

6 Results and Performance

6.1 Tested Flood Scenarios

Scenario	Rise rate	Peak level (m)	Final status
stable	0.0 m/s	0.50	monitoring
slow_rise	0.025	3.50	flood_risk
fast_rise	0.08	10.1	flood_risk
receding	-0.015	4.00	monitoring
flash_flood	0.15 (sigmoid)	3.45	flood_risk

Table 1: Performance across five representative wadi scenarios.

6.2 Monte Carlo Validation (20 runs)

- **Detection probability (P_d):** $\approx 90\%$
- **Falsealarm probability (P_{fa}):** $\approx 5\%$ under nominal noise
- The 60second persistence validator eliminated **every singletick false alarm**.

6.3 Latency

The complete decision pipeline from radar echo to alarm declaration runs in well under **1 second** on a standard CPU, supporting a 1 Hz update rate.

6.4 GUI Demonstration

A 7tab interactive MATLAB GUI shows live waterlevel trends, a 3D digital twin of the wadi with rising water, a semicircular risk gauge, an XAI featureimportance bar chart, and an event log. Screenshots are available in the project repository.

7 Impact

- **Saves lives:** 12 minutes of early warning enable evacuation before the flood peak.
- **Zerocarbon operation:** solarpowered, highenergy mode used $<5\%$ of the time.
- **Costeffective:** toolboxfree, runs on lowcost hardware, avoids billions in flood damage.
- **Closes Egypts earlywarning gap:** directly supports the national adaptation strategy and World Bank climate resilience funding.
- **Builds local expertise:** open, portable design empowers engineers and students to maintain and scale the system.

8 SDG Alignment

EcoPulse directly contributes to four United Nations Sustainable Development Goals:

Why: Remote wadis lack resilient monitoring infrastructure; commercial radars are too costly or powerhungry. **How EcoPulse contributes:** Delivers an opensource, toolboxfree radar system for lowcost hardware, promoting frugal innovation and climateresilient infrastructure.

Why: Continuous highpower operation wastes energy and shortens component life. **How EcoPulse contributes:** The dualmode green radar reduces power consumption by 95%, extending battery life and enabling solaronly operation with minimal electronic waste.

Why: Climate change is intensifying flash floods across Egypt and North Africa, demanding urgent adaptive measures. **How EcoPulse contributes:** Provides 12 minutes of accurate early warning, directly strengthening community resilience and reducing climaterelated disaster risk.

Why: Effective disaster resilience requires collaboration across technology, government, and humanitarian sectors. **How EcoPulse contributes:** Fully opensource and interoperable; integrates sensor networks, fuses local and meteorological data, and disseminates alerts via SMS, Telegram, and LoRa enabling multistakeholder partnerships for scalable early warning.

9 Scalability & Future Work

9.1 Beyond the Hackathon

The simulation is a complete digital twin all code is structured for direct porting to lowcost SDR hardware.

9.2 RealWorld Deployment Roadmap

1. **Phase 1:** Hardware prototype on Raspberry Pi + USRP with a 24 GHz frontend.
2. **Phase 2:** Single solarpowered pilot node in a highrisk Egyptian wadi (e.g., Wadi ElArish or Wadi Dahab).
3. **Phase 3:** Multinode mesh network integrated with Egypts national earlywarning system.

9.3 With More Time

- Train a deep LSTM on real Egyptian flood data for improved lead time.
- Design a ruggedised, dust and heatproof enclosure for extreme desert conditions.
- Implement full LoRaWAN mesh for widearea resilient alert dissemination.
- Obtain regulatory approval for 24 GHz spectrum use by civil protection agencies.

10 Challenges & Limitations

10.1 What We Faced

- Handcoding all algorithms (LSTM, fuzzy logic, Bayes SPRT, CFAR) without commercial toolboxes to ensure full Octave compatibility.
- Carefully tuning the 60second persistence window and multievidence fusion to eliminate false alarms without sacrificing sensitivity.

10.2 What Is Not Yet Solved

- Fully simulationbased realworld RF effects (phase noise, VCO drift, antenna coupling) have not been physically validated.
- Sitespecific calibration needed for each wadis unique geometry and local weather.
- Ruggedised enclosure for sandstorms, extreme heat, and humidity not yet designed.
- 24 GHz spectrum regulatory approval is pending.
- Scarce real flood data limits training of a deep LSTM.

11 Conclusion

EcoPulse delivers lifesaving flashflood warnings from a fully solarpowered, opensource 24 GHz radar designed for the worlds most remote and climatevulnerable wadis. Its dualmode green radar, multievidence AI fusion, and toolboxfree implementation prove that sustainable, autonomous early warning is achievable not as a future aspiration, but as a presentday simulationproven system ready for field deployment.
